

COVER PAGE

HR Analytics Dashboard

Employee Attrition & Workforce Insights using Microsoft Power BI

Prepared by: **Glory Chiamaka Emeh**

Data Analyst | Power BI | Excel | SQL | Python

Date: July 2026

Executive Summary

Employee attrition can have significant implications for recruitment costs, workforce productivity, employee morale, and organizational stability.

This project analyzes employee workforce data to identify patterns associated with employee attrition using Microsoft Excel for data preparation and Microsoft Power BI for interactive reporting.

The analysis identified notable differences in attrition across departments, age groups, job roles, overtime status, salary groups, and employee satisfaction indicators.

The resulting dashboard provides an interactive decision-support tool that enables HR stakeholders to explore workforce patterns, identify areas requiring further investigation, and support data-informed retention strategies.

Business Background

Human Resources departments continuously seek to understand why employees leave an organization.

High employee turnover results in:

- Increased recruitment costs
- Loss of institutional knowledge
- Reduced employee morale
- Lower organizational productivity
- Increased onboarding expenses

Understanding these factors enables organizations to design targeted interventions that improve employee retention and workforce stability.

Project Objectives

The objectives of this project were to:

- Analyze employee attrition trends.
- Evaluate workforce demographics.
- Measure employee satisfaction.
- Assess compensation patterns.
- Identify departments with high turnover.
- Examine overtime trends.
- Develop an interactive executive dashboard.
- Provide actionable business recommendations.

Dataset Overview

Item	Details
Dataset	HR Analytics Dataset
Source	Kaggle
Original Records	1,480
Cleaned Records	1,470
Duplicate Records Removed	10
Missing Values	57 in <code>YearsWithCurrManager</code>
Dashboard Tool	Microsoft Power BI
Cleaning Tool	Microsoft Excel

Methodology

The project followed the complete data analytics lifecycle.

Data Collection

The HR Analytics dataset was obtained from Kaggle.

Data Profiling

The dataset was assessed for duplicates, missing values, inconsistencies, and redundant fields.

Data Cleaning

The following tasks were completed:

- Removed duplicate employee records.
- Removed unnecessary columns.
- Corrected `BusinessTravel` inconsistencies.
- Standardized data formatting.
- Validated employee identifiers.

Data Modeling

The cleaned dataset was imported into Microsoft Power BI.

Calculated measures were developed using DAX to support executive reporting.

Dashboard Development

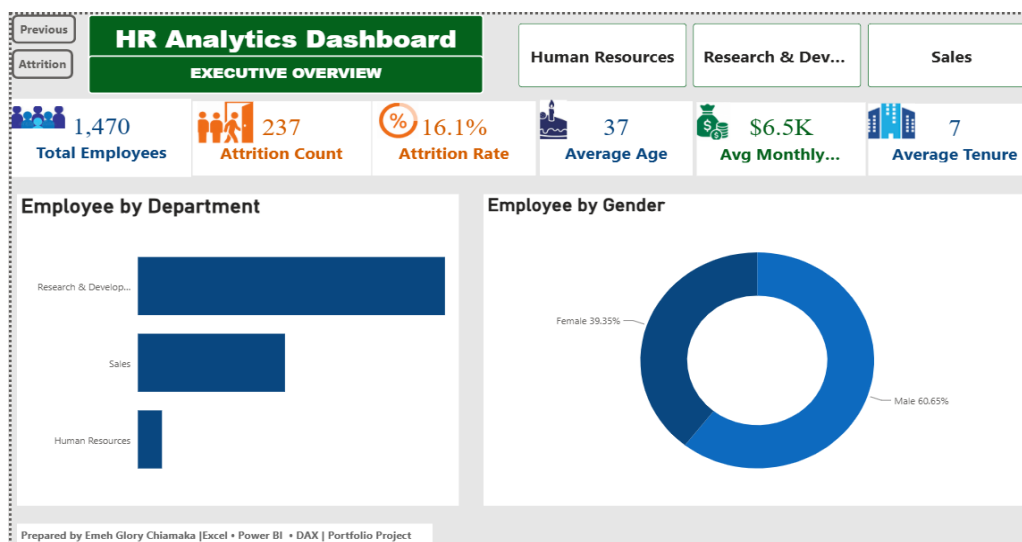
Interactive dashboards were designed to provide business users with meaningful workforce insights.

Dashboard Overview

The completed dashboard consists of five interactive pages.

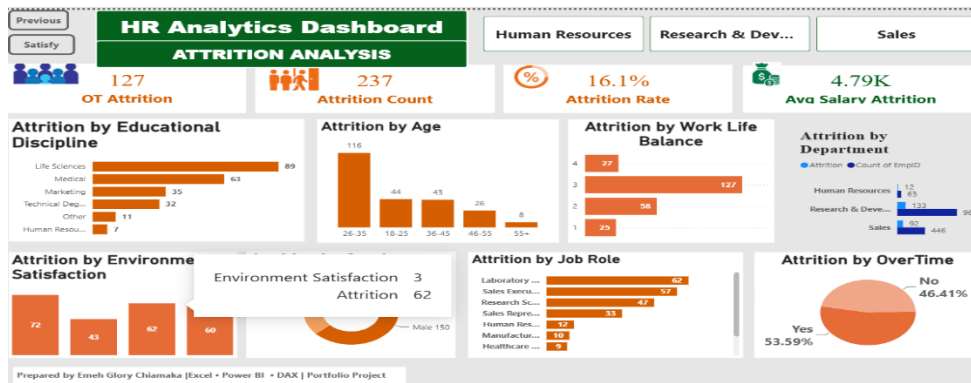
Executive Overview

Provides an overall summary of the workforce.



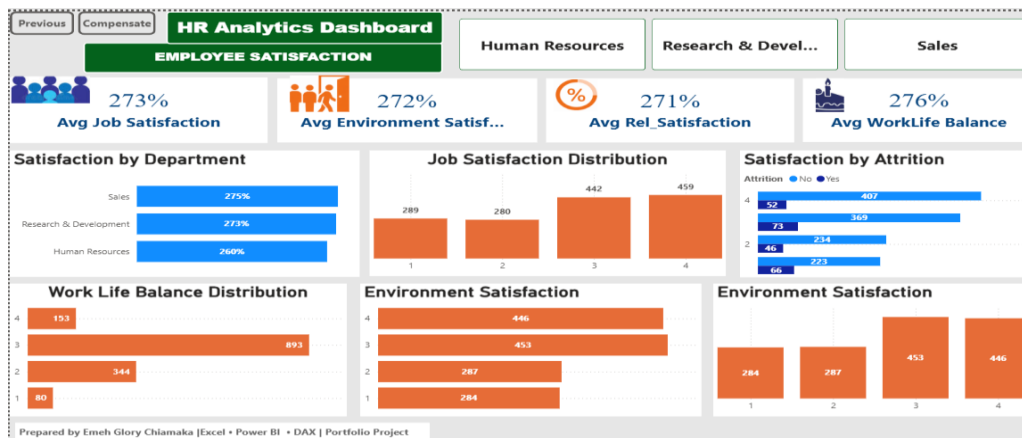
Attrition Analysis

Analyzes employee turnover across departments, age groups, overtime, gender, and education.



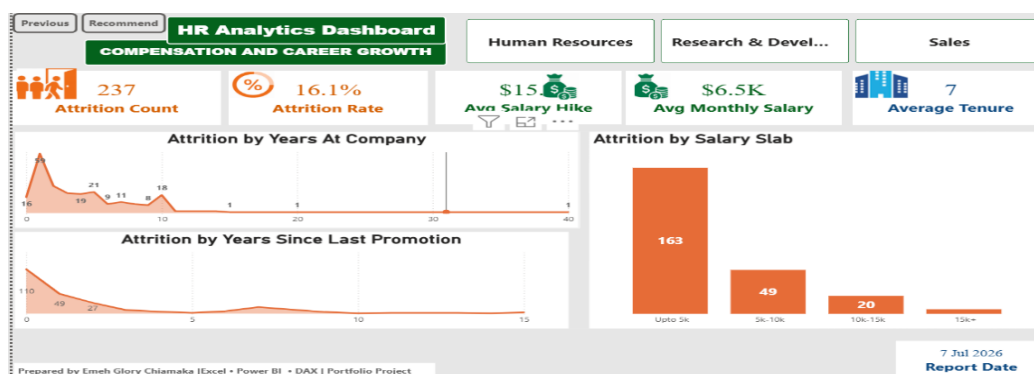
Employee Satisfaction

Measures satisfaction across multiple workforce dimensions.



Compensation & Career Growth

Analyzes salary, promotions, tenure, and employee growth.



Executive Recommendations

Summarizes actionable recommendations derived from the analysis.

HR RECOMMENDATIONS

High Priority

- Reduce excessive overtime.
- Investigate attrition among employees aged 26–35.

Medium Priority

- Improve retention for Laboratory Technicians.
- Review compensation for lower salary slabs.

Long-Term

- Strengthen career development and promotion pathways.

7 Jul 2026
Report Date

Key Findings

The analysis identified several important workforce trends.

1. Research & Development experienced the highest employee attrition.

This suggests that retention strategies should prioritize this department.

2. Employees aged 26–35 recorded the highest turnover.

Early-career professionals appear more likely to leave the organization.

3. Laboratory Technicians experienced the highest attrition.

This role may require improved engagement and career development initiatives.

4. Employees working overtime were significantly more likely to leave.

This indicates workload management should become a strategic HR priority.

5. Lower salary groups experienced higher attrition.

Compensation appears to influence employee retention.

6. Employee satisfaction strongly impacts retention.

Employees reporting lower satisfaction scores were more likely to leave the organization.

Business Recommendations

Based on the findings, the following recommendations are proposed.

Improve Work-Life Balance

Reduce excessive overtime through better workforce planning.

Strengthen Career Development

Introduce mentorship, learning opportunities, and structured promotion pathways.

Review Compensation Strategy

Benchmark salary structures against industry standards.

Increase Employee Engagement

Conduct regular employee satisfaction surveys.

Focus Retention Programs

Develop targeted retention initiatives for departments and job roles experiencing high attrition.

Expected Business Impact

Implementing these recommendations could help the organization:

- Reduce employee turnover.
- Lower recruitment costs.
- Improve employee satisfaction.
- Increase workforce productivity.
- Strengthen organizational stability.
- Support data-driven HR decision-making.

Limitations

This analysis identifies patterns and associations within the available dataset but does not establish causal relationships between employee characteristics and attrition.

The dataset contains 57 missing values in `YearsWithCurrManager`, which may affect analyses involving tenure with the current manager.

The analysis is also based on a single HR dataset. Additional information such as exit interview responses, absenteeism, performance history, employee engagement surveys, recruitment costs, and historical workforce data could provide deeper insight into the reasons behind employee turnover.

Therefore, the findings should be used as a basis for further HR investigation and decision-making rather than as evidence that any single factor directly causes employee attrition.

Conclusion

This project demonstrates how data analytics can transform workforce data into actionable business intelligence.

By combining Microsoft Excel, DAX, and Microsoft Power BI, the dashboard enables HR professionals to identify workforce trends, monitor employee engagement, and develop proactive retention strategies.

The project highlights the value of data-driven decision-making in improving organizational performance.

About the Analyst

Glory Chiamaka Emeh

Data Analyst specializing in:

- Microsoft Power BI
- Microsoft Excel
- SQL
- Python
- Business Intelligence
- Data Visualization
- Dashboard Development

GitHub:

<https://github.com/emehglorychiamaka>